

Exam Date & Time: 19-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester Makeup Examination, June -2025

Agile Software Development for Engineers [IT-4308]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) In what ways does the Kanban system assist in managing and optimizing workflow within a team? Furthermore, what role do Work in Progress (WIP) limits play in this process? (5)
 - A)
 - B) In what ways can virtual meetings enhance the evaluation of Agile methodologies in distributed startup teams using Scrum methodology. (5)
- 2) How can Agile teams strike a balance between delivering working software and maintaining appropriate documentation? (5)
 - A)
 - B) Evaluate the effectiveness of Agile scaling frameworks such as SAFe (Scaled Agile Framework) or LeSS (Large-Scale Scrum) in addressing the challenges of coordinating multiple Agile teams within large organizations. (5)
- 3) Illustrate the process of scrum framework with an example. (5)
 - A)
 - B) Illustrate the possible combinations of pair programming in software development life cycle. (5)
- 4) Critically evaluate two Agile project management tools (e.g., Jira vs. Azure DevOps) in terms of scalability and team collaboration. (5)
 - A)
 - B) Design a stakeholder engagement plan for a product with evolving requirements and multiple end-user groups. How will you gather feedback and incorporate it iteratively? (5)
- 5) Discuss the influence of DevOps adoption on Agile delivery performance. Which key performance indicators (KPIs) reflect the success or failure of this integration, and why? (5)
 - A)
 - B) Describe your approach to coordinating sprint planning and backlog grooming when several Agile teams depend on a shared product backlog. How do you ensure strategic

alignment and minimize cross-team conflicts?

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